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At the beginning of this lab, machine learning felt very abstract to me. I understood the basic definitions of supervised, unsupervised, and reinforcement learning, but I did not fully grasp how they were actually applied in practice. One of the most confusing parts at first was understanding why we had to split data into training and testing sets instead of just using all the data. As I worked through the lab and saw how model performance was evaluated on unseen data, it clicked that this step is critical to avoid misleading results and to better understand how a model might behave in the real world.

Seeing the full machine learning workflow laid out step by step helped connect concepts that previously felt disconnected. Data preparation, model training, and evaluation are not isolated tasks each step directly affects the next. For example, I realized that the quality of the input data plays a huge role in how well the model performs, sometimes even more than the choice of algorithm. This helped me understand why machine learning is not just about choosing a model, but about carefully handling data throughout the entire process.

The idea of validation and overfitting also became much clearer during this lab. Before, overfitting sounded like a theoretical problem, but seeing how a model can perform very well on training data and worse on test data made it feel real. It showed me why evaluation metrics and validation matter, especially when comparing models. Understanding this helped me better interpret accuracy scores and confusion matrices instead of just looking at which number was higher.

This lab also helped me better distinguish between supervised, unsupervised, and reinforcement learning in practical terms. Supervised learning now makes sense to me as learning from labeled examples, like the wine classification task. Unsupervised learning is about finding patterns when labels are not available, and reinforcement learning focuses on learning through trial and error using rewards. Being able to connect each learning type to real-world examples made these distinctions much clearer.

Overall, this lab helped shift my understanding of machine learning from a collection of definitions into a structured process. I now see machine learning as an iterative workflow where data, models, and evaluation all work together. While I still have a lot to learn, this exercise gave me a much stronger foundation and made the concepts feel more approachable and practical.